



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING(24CI3201)

ASE Course Project

A.Y. 2026 - 2027

TITLE OF THE PROJECT		
S. No.	University ID	Name
1	2420030531	Vuppuluri Harshita
2	2420030469	Luhitha Boppana
3	2420030399	Cheekoti Manognya
Name of the Guide		G. Lavanya

PROJECT TITLE

Adaptive AI Learning Assistant for Personalized College Student Learning

ABSTRACT

College students have different learning abilities, study patterns, and academic requirements, making a common learning approach less effective. Existing learning platforms mainly provide static study materials and assessments without continuously adapting to individual student performance. This creates challenges in identifying weak subjects, maintaining consistent study schedules, and providing appropriate learning support.

The proposed Adaptive AI Learning Assistant addresses these challenges by providing an intelligent and personalized learning platform for college students. The system analyzes quiz performance, study time, attendance, subject-wise progress, and learning patterns to identify individual learning gaps. It uses Artificial Intelligence (AI), Machine Learning (ML), and Large Language Models (LLMs) to generate personalized study plans, adaptive quizzes, and simplified explanations. The system continuously updates recommendations based on student performance and interactions, enabling adaptive learning support.

Research Gap: Existing educational platforms generally provide standardized learning content and basic performance tracking but lack continuous personalization, adaptive quiz generation, intelligent study-plan recommendations, and performance prediction based on multiple student learning parameters. Most systems require students to manually identify their weak areas and plan their learning activities.

Software and Technologies Used: React.js, Node.js, Express.js, MongoDB, Google Gemini API, Scikit-learn, REST APIs, JWT Authentication, Git & GitHub, Docker, and Postman.

Expected Outcome: The proposed system will help students identify weak subjects, receive personalized study plans, practice adaptive quizzes, obtain AI-based explanations, and monitor their academic progress. The platform aims to improve learning efficiency, student engagement, and academic performance through continuous AI-driven personalization and adaptive software engineering principles.

Signature of the Guide

HOD